

Suleyman Hajizadeh

BIOINFORMATICIAN

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📍 Baku, Azerbaijan

CORE INTERESTS & EXPERTISE

Genomics & transcriptomics · Sequence alignment & variant profiling · High-throughput gene expression analysis · Statistical data analysis & visualization · Workflow automation (Snakemake) · Structural bioinformatics

EDUCATION

Western Caspian University

Baku, Azerbaijan · 2022 – Jun 2026

Bachelor of Science in Biology

- ▶ Relevant coursework: Molecular Biology, Genetics, Biochemistry, Biostatistics, Computational Methods in Biology
- ▶ Independent research in computational cancer genomics and transcriptomics conducted in parallel with degree coursework

PROJECTS & RESEARCH EXPERIENCE

High-Throughput RNA-Seq & Gene Expression Pipeline — TNBC

2024 – Present

Institute of Molecular Biology & Biotechnology (IMBB), Baku

- ▶ Developed a **Snakemake** workflow (SRA → FastQC → Trimmomatic → HISAT2/GRCh38 → featureCounts → DESeq2) for fully reproducible bulk RNA-seq data analysis of GSE183947 (TNBC cohort)
- ▶ Implemented **DESeq2** differential expression with statistical commentary: derived negative binomial model assumptions, BH-FDR correction mathematics, and VST vs. rlog trade-offs in a dedicated *Statistical Methods* primer
- ▶ Designed a nested Leave-One-Out Cross-Validation classifier with feature selection inside training folds to prevent data leakage; permutation test empirical p-value < 0.05

Custom Sequence Alignment & Phylogenomics CLI

2025

- ▶ Built **phylo_pipeline.py** — a **pure Python** command-line tool that fetches CDS sequences from NCBI Entrez, performs pairwise alignment, and computes **K2P** and **JC69 distance matrices from mathematical formulas**
- ▶ Implemented the **Neighbor-Joining algorithm from scratch** in NumPy; reconstructed the evolutionary history of COL1A1 across 7 vertebrate species (*H. sapiens*, *Pan troglodytes*, *M. musculus*, *R. norvegicus*, *B. taurus*, *G. gallus*, *D. rerio*)
- ▶ Outputs: Newick tree file (.nwk) and high-resolution cladogram PNG — **no MEGA GUI dependency**

Protein 3D Structural Biophysics & Coordinate Analyzer2025

- ▶ Parsed crystallographic PDB coordinate files with Biopython; extracted Cα atoms and computed 3D Euclidean distance matrices and backbone RMSD between structural domains
- ▶ Implemented **hydrophobic core detection** using Kyte-Doolittle scale weighted by spatial packing density (8 Å threshold)
- ▶ Coded a **Miyazawa-Jernigan statistical contact potential** loop to estimate non-covalent folding free energies at residue-pair resolution
- ▶ Computed Pearson correlation between Cα packing density and crystallographic B-factors ($r = -0.37$ for *insulin 1COH*), mathematically validating core rigidity

AlphaFold2 Structural Analysis — AKT1 Kinase (TNBC)2024

- ▶ Predicted 3D structure of AKT1 hub kinase using ColabFold v1.6.1 (AlphaFold2 multimer); mean pLDDT = 65.61 across full chain
- ▶ Generated per-residue pLDDT confidence plots, Ramachandran backbone torsion analysis, Cα contact maps, and PyMOL publication-quality renders

TECHNICAL SKILLS

Languages	Python (Advanced) · R (Intermediate) · Bash/Shell scripting
Genomics & Align.	HISAT2 · featureCounts · Trimmomatic · FastQC · BLAST · SRA Toolkit · Clustal/Muscle
R / Bioconductor	DESeq2 · WGCNA · edgeR · pROC · ggplot2 · limma
Sci. Python	Biopython · NumPy · Pandas · SciPy · Scikit-Learn · Matplotlib
Pipelines	Snakemake · Conda · Git/GitHub
Structural Bio	AlphaFold2 (ColabFold) · PyMOL · PDB parsing · RMSD · contact maps
Statistics / ML	Negative Binomial model · BH-FDR · Nested CV · Random Forest · PCA · GSEA

AFFILIATIONS & TRAINING

Institute of Molecular Biology & Biotechnology (IMBB)Baku, Azerbaijan · 2023 – Present
Bioinformatician (independent researcher)

Data Analysis for Life SciencesHarvard University · edX · Expected Jul 2026

Bioinformatics MicroMastersUniversity of Maryland Global Campus · edX · Expected Aug 2026

LANGUAGES

Azerbaijani	English	Russian
Native	Professional (IELTS target 7.5+, Jul 2026)	Conversational